

In this thesis, we designed an interactive generative tool based on the Variational Autoencoder (VAE) to synthesise new and interesting drum one-shot samples for electronic music production. We researched audio representations used in audio generation tasks and selected two that suited our settings the best. Furthermore, we created our own dataset of almost 16,000 freely available samples organised into 9 drum categories. Through a series of experiments, we were able to achieve a model that was reconstructing and generating quality samples; however, they contained a specific noise artefact we could not get rid of. Based on the results of PCA and convex combinations methods we found out that the latent space has “meaningful” properties.